

**UNIVERSITY OF THE FREE STATE
UNIVERSITEIT VAN DIE VRYSTAAT**

BLOEMFONTEIN CAMPUS/KAMPUS

CSIS3744

**DEPARTMENT: COMPUTER SCIENCE AND INFORMATICS
DEPARTEMENT: REKENAARWETENSKAP EN INFORMATIKA**

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**EXAMINATION/EKSAMEN:
Additional End-of-year/Bykomende Jaareind 2018
PAPER/VRAESTEL 1**

ASSESSOR(S)/

ASSESSOR(E): 1. Mnr. W.S.J. Marais

MODERATOR(S)/

MODERATOR(E): 1. Dr. B.P. Haskins

TIME/TYD: 180 min

MARKS/PUNTE: 100

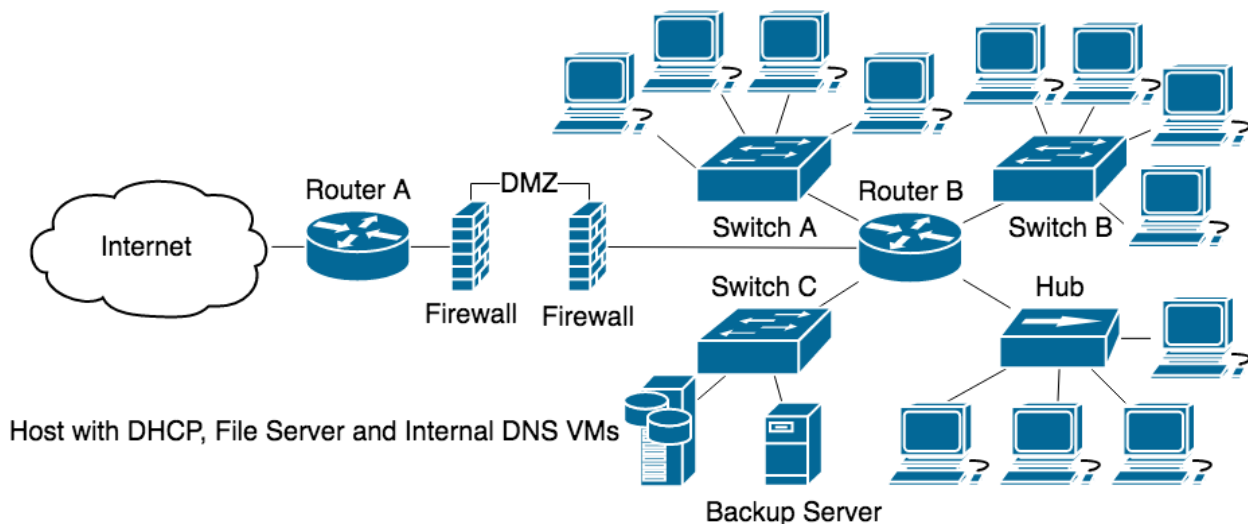
INSTRUCTIONS: Answer all the questions. Number your questions clearly and correctly. The Afrikaans versions of the questions are in italics.

INSTRUKSIES: *Beantwoord al die vrae. Nommer u antwoorde duidelik en korrek. Die Afrikaanse weergawes van die vrae is kursief gedruk.*

Question / Vraag 1 [29 Marks / Punte]

The following is a network diagram of a client/server MAN that belongs to Z-Star Technologies. The network implements the 100BaseT Ethernet standard: /

Die volgende is 'n netwerkdiagram van 'n kliënt/bediener-MAN wat aan Z-Star Technologies behoort. Die netwerk implementeer die 100BaseT Ethernet-standaard:



- 1.1 Explain why Z-Star Technologies' network forms a MAN? /
Verduidelik waarom Z-Star Technologies se netwerk 'n MAN vorm? (2)
- 1.2 Specify how many collision domains and how many broadcast domains the current LAN contains. /
Spesifiseer hoeveel botsing-domeine en hoeveel uitsaai-domeine die huidige LAN bevat. (2)
- 1.3 Specify the router type (Interior, exterior or border) for router A and router B. /
Spesifiseer die tipe roetebepaler (Interne, eksterne of grens) vir roetebepaler-A en roetebepaler-B. (2)
- 1.3.1 Discuss the similarities and differences between the IS-IS and OSPF routing protocols. /
Bespreek die ooreenkomste en verskille tussen die IS-IS- en OSPF-roetebepalingsprotokolle. (4)
- 1.3.2 Which of the two routers (A and/or B) will implement the BGP routing protocol? /
Watter van die twee roetebepalers (A en/of B) sal die BGP-roetebepalingsprotokol implementeer? (1)
- 1.4 Z-Star Technologies would like to upgrade their network backbone to conform to the 1000Base-LX Ethernet standard using multimode fibre (MMF). /
Z-Star Technologies wil hul netwerkruggraad opgradeer om aan die 1000Base-LX Ethernet-standaard te voldoen deur multimode optiese-vesel (MMF) te gebruik.
- 1.4.1 Explain the meaning of the Ethernet standard that Z-Star Technologies want to implement. /
Verduidelik die betekenis van die Ethernet-standaard wat Z-Star Technologies wil implementeer. (3)
- 1.4.2 What is the maximum segment length for the intended Ethernet standard? /
Wat is die maksimum segmentlengte vir die beoogde Ethernet-standaard? (1)
- 1.4.3 Briefly discuss two factors that influence the maximum segment length of the specified Ethernet standard. /
Bespreek kortliks twee faktore wat die maksimum segmentlengte beïnvloed van die gespesifiseerde Ethernet-standaard. (2)
- 1.4.4 Indicate which network components will have to be replaced in order to meet the requirements of the intended Ethernet standard and specify what these components will have to be replaced with. /
Dui aan watter netwerkkomponente vervang moet word om aan die vereistes van die beoogde Ethernet-standaard te voldoen en spesifiseer waarmee hierdie komponente vervang moet word. (5)
- 1.4.5 Specify how many collision domains and how many broadcast domains the intended MAN will contain. /
Spesifiseer hoeveel botsing-domeine en hoeveel uitsaai-domeine die beoogde MAN sal bevat. (2)

- 1.5 Explain the difference between incremental and differential backups as carried out by the backup server. /
Verduidelik die verskil tussen inkrementele en differensiële rugsteunkopieë wat deur die rugsteunbediener uitgevoer word. (2)
- 1.6 Discuss the purpose of the demilitarised zone (DMZ) and give two possible types of servers you would place in the DMZ. /
Bespreek die doel van die gedemilitariseerde sone (DMZ) en gee twee moontlike tipes bedieners wat u in die DMZ sal plaas. (3)
 [29]

Question / Vraag 2 [22 Marks / Punte]

The users on Z-Star Technologies' network requested that the network be upgraded to allow them to roam around the building with their laptops and mobile devices without losing connectivity. /

Die gebruikers op die netwerk van Z-Star Technologies het gevra dat die netwerk opgegradeer word sodat hulle deur die gebou met hul skootrekenaars en mobiele toestelle kan rondbeweeg sonder om konnektiwiteit te verloor.

- 2.1 Describe how a wireless network can be set up to allow the users to roam around the entire building. /
Beskryf hoe 'n draadlose netwerk opgestel kan word om die gebruikers in staat te stel om deur die hele gebou rond te beweeg. (3)
- 2.2 Compare all the Wi-Fi standards in table format with regard to frequency band and maximum theoretical throughput. /
Vergelyk al die Wi-Fi standaarde in tabelformaat ten opsigte van frekwensieband en maksimum teoretiese deursetvermoë. (7)
- 2.3 Name and discuss the purpose of the three different wireless frame groups. /
Noem en bespreek die doel van die drie verskillende draadlose raamgroepe. (6)
- 2.4 Discuss the differences between the header information of an 802.11 data frame and an 802.3 frame with regard to their MAC addresses. /
Bespreek die verskille tussen die kopstukinligting van 'n 802.11-dataraam en 'n 802.3-raam ten opsigte van hul MAC-adresse. (6)
 [22]

Question / Vraag 3 [21 Marks / Punte]

The internal network address of Z-Star Technologies is 172.125.0.0. They want to create two separate subnets on their network. /

Die interne netwerkadres van Z-Star Technologies is 172.125.0.0. Hulle wil twee afsonderlike subnetwerke op hul netwerk te skep.

- 3.1 Calculate the new subnet mask for Z-Star Technologies' subnets. /
Bereken die nuwe subnetmasker vir Z-Star Technologies se subnetwerke. (4)

3.2 How many usable addresses are available per subnet? /
Hoeveel bruikbare adresse is beskikbaar per subnetwork? (1)

3.3 Complete the table below by supplying each subnet's network address, broadcast address and usable host address range: /
Voltooi die onderstaande tabel deur elke subnetwork se netwerkadres, uitsaai-adres en bruikbare gasheer-adresreeks te verskaf:

	Subnet	Network Address <i>Netwerkadres</i>	Broadcast Address <i>Uitsaai-adres</i>	Usable Host Address Range <i>Bruikbare Gasheer-adresreeks</i>	
3.3.1	1				(2)
3.3.2	2				(2)

3.4 Give five reasons for using VLANs on a network. /
Gee vyf redes vir die gebruik van VLAN's op 'n netwerk. (5)

3.4.1 After studying the network diagram of Z-Star Technologies, discuss whether VLANs would contribute to the effectiveness their network. /
Nadat u die netwerkdiagram van Z-Star Technologies bestudeer het, bespreek of VLAN's tot die doeltreffendheid van hul netwerk sal bydra. (3)

3.5 Explain how trunking protocols operate.
Verduidelik hoe stamprotokolle funksioneer. (2)

3.5.1 Why is Cisco's VLAN trunking protocol the most popular trunking protocol? /
Hoekom is Cisco se VLAN-stamprotokol die gewildste stamprotokol? (2)
[21]

Question / Vraag 4 [14 Marks / Punte]

Study the following header and payload information of an intercepted network packet. The information has already been translated into readable text format. /
Bestudeer die volgende kopstuk- en "payload"-inligting van 'n onderskepte netwerkpakkie. Die inligting is reeds in leesbare teksformaat vertaal.

Bit 0-7		Bit 8-15		Bit 16-23		Bit 24-31	
4	20 bytes	0		84 bytes			
0				1	0	0	
252		ICMP		fa30			
192.168.1.2							
192.168.2.11							
Echo request		0		45da			

4.1 Which Network layer protocols were used during the transmission of the packet? /
Watter Netwerkvlaakprotokols is tydens die oordrag van die pakkie gebruik? (2)

4.2 What is the total size of the packet? /
Wat is die totale grootte van die pakkie? (1)

- 4.3 Provide the source and destination IP addresses for the transmission. /
Verskaf die bron- en bestemming IP-adresse vir die oordrag. (2)
- 4.4 How many hops can the packet still be forwarded before it will be discarded? /
Hoeveel "hops" kan die pakkie nog steeds aangestuur word voordat dit weggegooi word? (1)
- 4.5 Determine the type and purpose of the packet that was transmitted. /
Bepaal die tipe en die doel van die pakkie wat oorgedra is. (2)
- 4.6 Draw an Ethernet II frame and label the different components of the frame. /
Teken 'n Ethernet II raam en benoem die verskillende komponente van die raam. (6)
[14]

Question / Vraag 5 [14 Marks / Punte]

The DNS server, ns1.server.co.za, has the following IP address: 10.1.1.131. Its IP address was assigned by a DHCP server. /
Die DNS-bediener, ns1.server.co.za, het die volgende IP-adres: 10.1.1.131. Sy IP-adres is deur 'n DHCP-bediener toegeken.

- 5.1 Discuss the steps involved in the DHCP leasing process. /
Bespreek die stappe wat betrokke is met die DHCP verhuringsproses. (4)
- 5.2 Write down the socket address that is used to request DNS information from the server. /
Skryf die sokadres neer wat gebruik word om DNS inligting van die bediener af aan te vra. (1)
- 5.3 What console command can be used to determine whether the DNS server is responding? Write down the full command. /
Watter konsole-instruksie kan gebruik word om te bepaal of die DNS-bediener reageer? Skryf die volledige instruksie neer. (2)
- 5.3.1 Explain how this command uses ICMP and ARP to perform its task. /
Verduidelik hoe hierdie instruksie ICMP en ARP gebruik om sy taak uit te voer. (3)
- 5.4 What domain does the DNS server belong to? /
Aan watter domein behoort die DNS-bediener? (1)
- 5.5 What is the host name of the DNS server? /
Wat is die gasheernaam van die DNS-bediener? (1)
- 5.6 Briefly explain the difference between unicast, and multicast IPv6 addresses. /
Verduidelik kortliks die verskil tussen "unicast", en "multicast" IPv6-adresse. (2)
[14]

End of Paper / Einde van Vraestel